Networking Fundamentals

Introduction to Networks

What is a Network?

- A `network` is a collection of interconnected devices that can communicate , sending and receiving data, and also sharing resources with each other.
- These individual endpoint devices, often called `nodes`, include computers, smartphones, printers, and servers.

Concepts	Description
Nodes	Individual devices connected to a network.
Links	Communication pathways that connect nodes (wired or wireless)
Data Sharing	The primary purpose of a network is to enable data exchange.

Why Are Networks Important?

Networks, particularly since the advent of the Internet, have radically transformed society, enabling the multitude of possibilities that are now essential to our lives. Below are just a few of the benefits afforded to us by this incredible technology.

Function	Description
Resource Sharing	Multiple devices can share hardware (like printers) and software resources.
Communication	Instant messaging, emails, and video calls rely on networks.
Data Access	Access files and databases from any connected device.
Collaboration	Work together in real-time, even when miles apart.

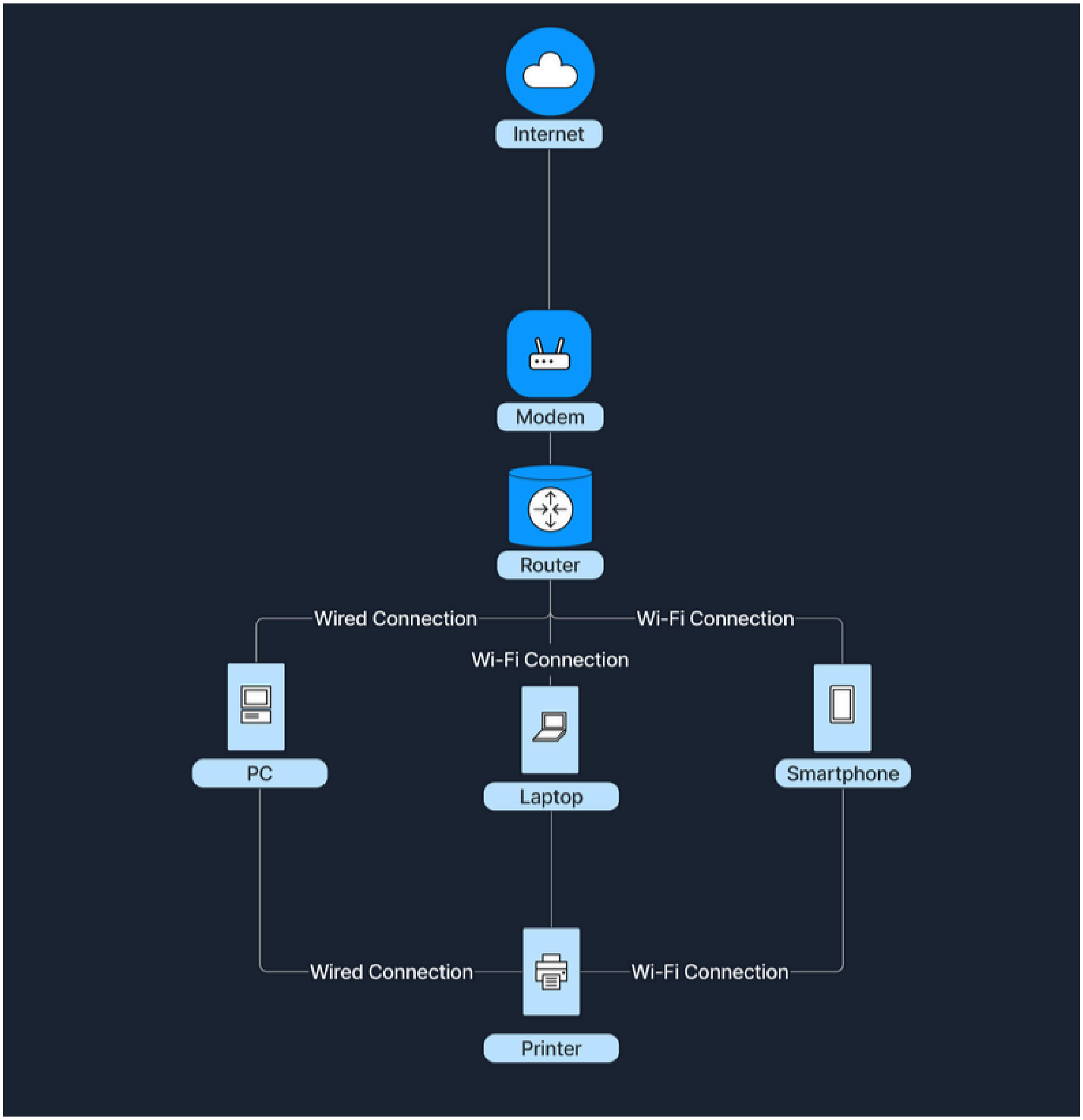
Types of Networks

Networks vary in size and scope. The two primary types are `Local Area Network (LAN)` and `Wide Area Network (WAN)`.

Local Area Network (LAN)

A `Local Area Network (LAN)` connects devices over a short distance, such as within a home, school, or small office building. Here are some of its key characteristics:

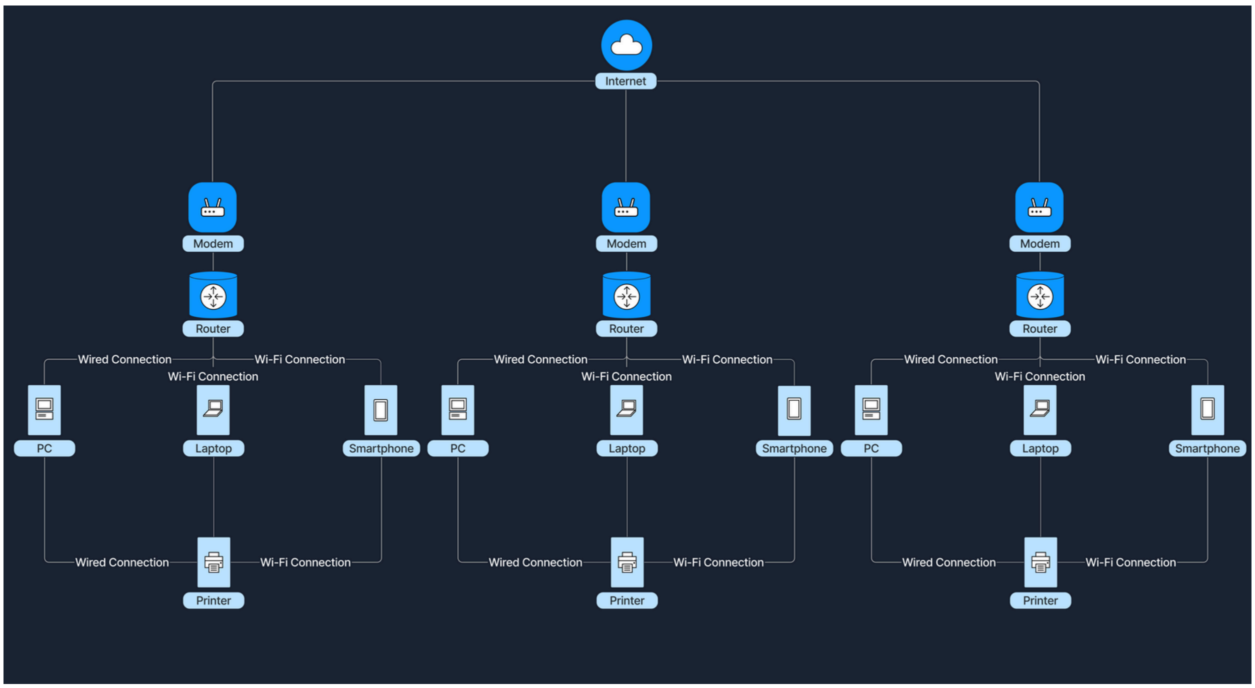
Characteristic	Description
Geographical Scope	Covers a small area.
Ownership	Typically owned and managed by a single person or organization.
Speed	High data transfer rates.
Media	Uses wired (Ethernet cables) or wireless (Wi-Fi) connections.



Wide Area Network (WAN)

A `Wide Area Network (WAN)` spans a large geographical area, connecting multiple LANs. Below are some of its key characteristics:

Characteristic	Description
Geographical Scope	Covers cities, countries, or continents.
Ownership	Often a collective or distributed ownership (e.g., Internet Service Providers).
Speed	Slower data transfer rates compared to LANs due to long-distance data travel.
Media	Utilizes fiber optics, satellite links, and leased telecommunication lines.



How Do LANs and WANs Work Together?

- A LAN (Local Area Network) can connect to a WAN (Wide Area Network) to communicate with networks outside its local area.
- At home, devices connect to a router (LAN), and the router connects to the ISP's WAN through a modem.
- The modem converts digital signals so they can travel through telephone, cable, or fiber-optic lines.

- This connection allows devices on the LAN to access the Internet and online services worldwide.
- In businesses, multiple office LANs are connected through a WAN, enabling employees to share data, access centralized resources, and collaborate across locations.

Devices → Router (LAN) → Modem → ISP WAN → Internet

Network Concepts

OSI Model

The `Open Systems Interconnection (OSI) model` is a conceptual framework that standardizes the functions of a telecommunication or computing system into seven abstract layers. This model helps vendors and developers create interoperable network devices and software. Below we see the seven layers of the OSI Model.

Physical Layer (Layer 1)

The `Physical Layer` is the first and lowest layer of the OSI model. It is responsible for transmitting raw bitstreams over a physical medium. This layer deals with the physical connection between devices, including the hardware components like Ethernet cables, hubs, and repeaters.

Data Link Layer (Layer 2)

The `Data Link Layer` provides node-to-node data transfer - a direct link between two physically connected nodes. It ensures that data frames are transmitted with proper synchronization, error detection, and correction. Devices such as switches and bridges operate at this layer, using MAC (Media Access Control) addresses to identify network devices.

Network Layer (Layer 3)

The `Network Layer` handles packet forwarding, including the routing of packets through different routers to reach the destination network. It is responsible for logical addressing and path determination, ensuring that data reaches the correct destination across multiple networks. Routers operate at this layer, using IP (Internet Protocol) addresses to identify devices and determine the most efficient path for data transmission.

Transport Layer (Layer 4)

The `Transport Layer` provides end-to-end communication services for applications. It is responsible for the reliable (or unreliable) delivery of data, segmentation, reassembly of messages, flow control, and error checking. Protocols like `TCP (Transmission Control Protocol)` and `UDP (User Datagram Protocol)` function at this layer. TCP offers reliable, connection-oriented transmission with error recovery, while UDP provides faster, connectionless communication without guaranteed delivery.

Session Layer (Layer 5)

The `Session Layer` manages sessions between applications. It establishes, maintains, and terminates connections, allowing devices to hold ongoing communications known as sessions. This layer is essential for session checkpointing and recovery, ensuring that data transfer can resume seamlessly after interruptions. Protocols and `APIs (Application Programming Interfaces)` operating at this layer coordinate communication between systems and applications.

Presentation Layer (Layer 6)

The `Presentation Layer` acts as a translator between the application layer and the network format. It handles data representation, ensuring that information sent by the application layer of one system is readable by the application layer of another. This includes data encryption and decryption, data compression, and converting data formats. Encryption protocols and data compression techniques operate at this layer to secure and optimize data transmission.

Application Layer (Layer 7)

The `Application Layer` is the topmost layer of the OSI model and provides network services directly to end-user applications. It enables resource sharing, remote file access, and other network services. Common protocols operating at this layer include `HTTP (Hypertext Transfer Protocol)` for web browsing, `FTP (File Transfer Protocol)` for file transfers, `SMTP (Simple Mail Transfer Protocol)` for email transmission, and `DNS (Domain Name System)` for resolving domain names to IP addresses. This layer serves as the interface between the network and the application software.

Layer 7 (Application)

- HTTP = HyperText Transfer Protocol
- HTTPS = HyperText Transfer Protocol Secure
- FTP = File Transfer Protocol
- SMTP = Simple Mail Transfer Protocol
- DNS = Domain Name System

Layer 6 (Presentation)

- SSL = Secure Sockets Layer
- TLS = Transport Layer Security

Layer 5 (Session)

- API = Application Programming Interface

Layer 4 (Transport)

- TCP = Transmission Control Protocol
- UDP = User Datagram Protocol

Layer 3 (Network)

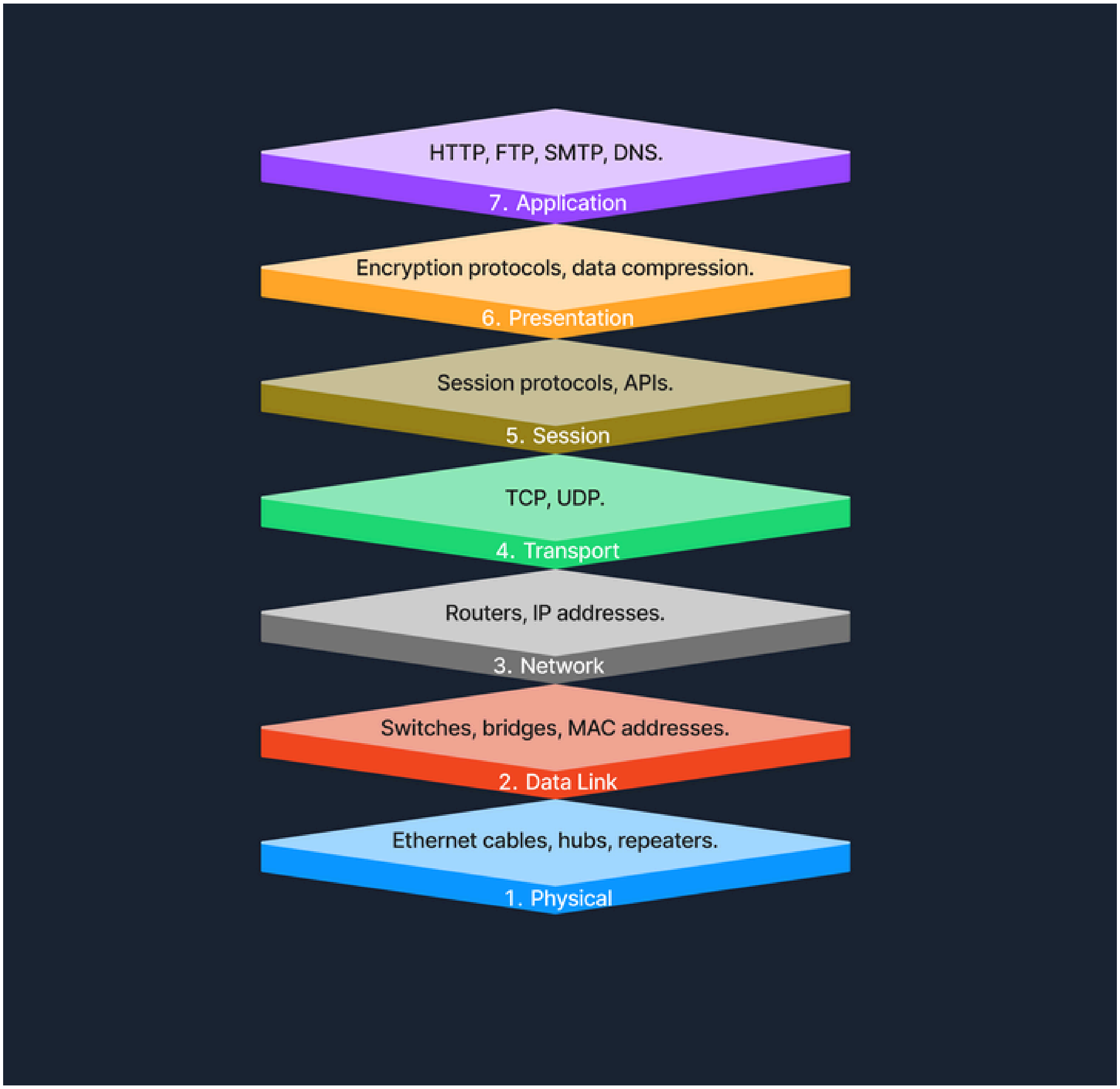
- IP = Internet Protocol
- ICMP = Internet Control Message Protocol
- ARP = Address Resolution Protocol

Layer 2 (Data Link)

- MAC = Media Access Control

Layer 1 (Physical)

- NIC = Network Interface Card



TCP/IP Model

The `Transmission Control Protocol/Internet Protocol (TCP/IP) model` is a condensed version of the `OSI` model, tailored for practical implementation on the internet and other networks. Below we see the four layers of the `TCP/IP Model`.

Link Layer

This layer is responsible for handling the physical aspects of network hardware and media. It includes technologies such as Ethernet for wired connections and Wi-Fi for wireless connections. The Link Layer corresponds to the Physical and Data Link Layers of the OSI model, covering everything from the physical connection to data framing.

Internet Layer

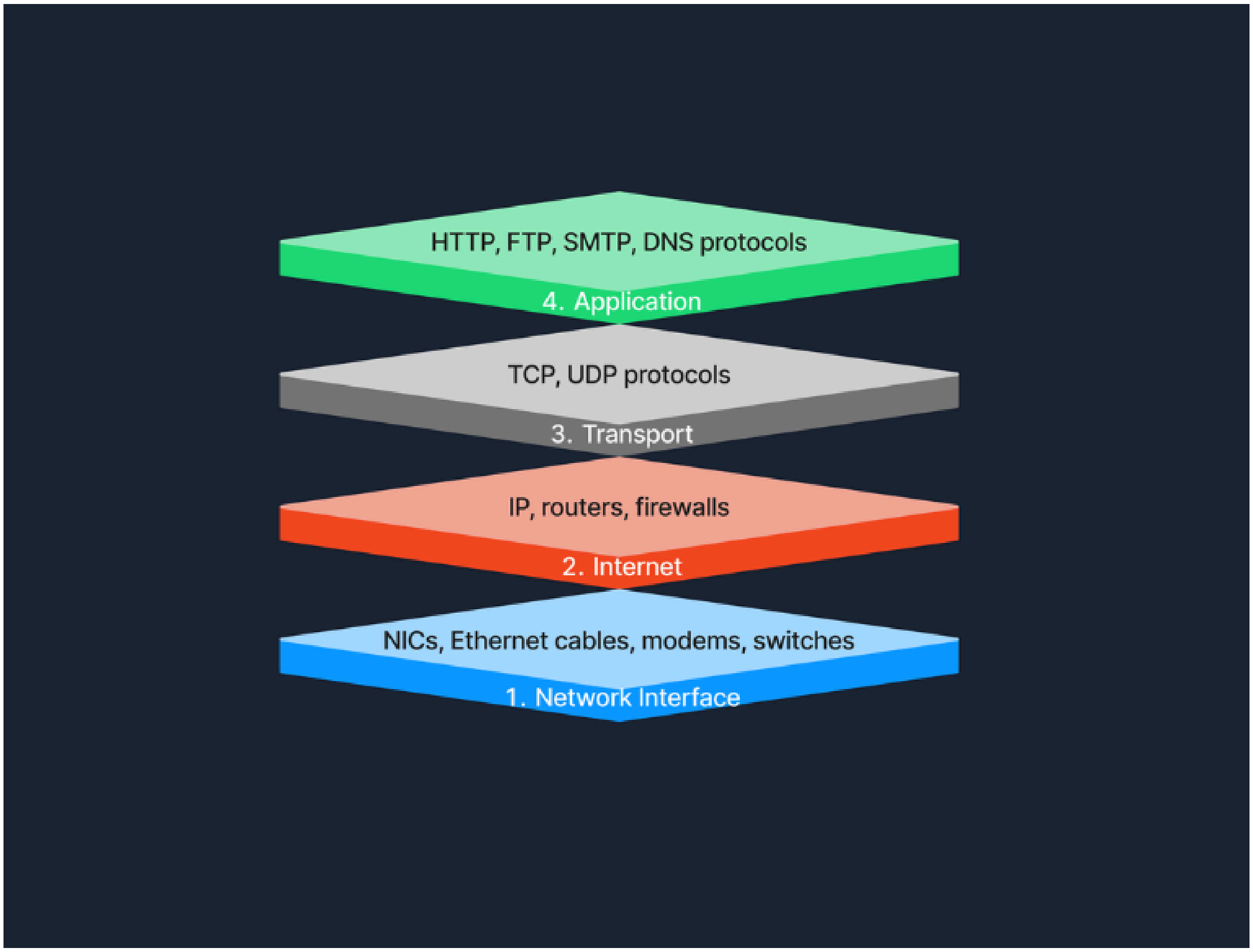
The `Internet Layer` manages the logical addressing of devices and the routing of packets across networks. Protocols like IP (Internet Protocol) and ICMP (Internet Control Message Protocol) operate at this layer, ensuring that data reaches its intended destination by determining logical paths for packet transmission. This layer corresponds to the Network Layer in the OSI model.

Transport Layer

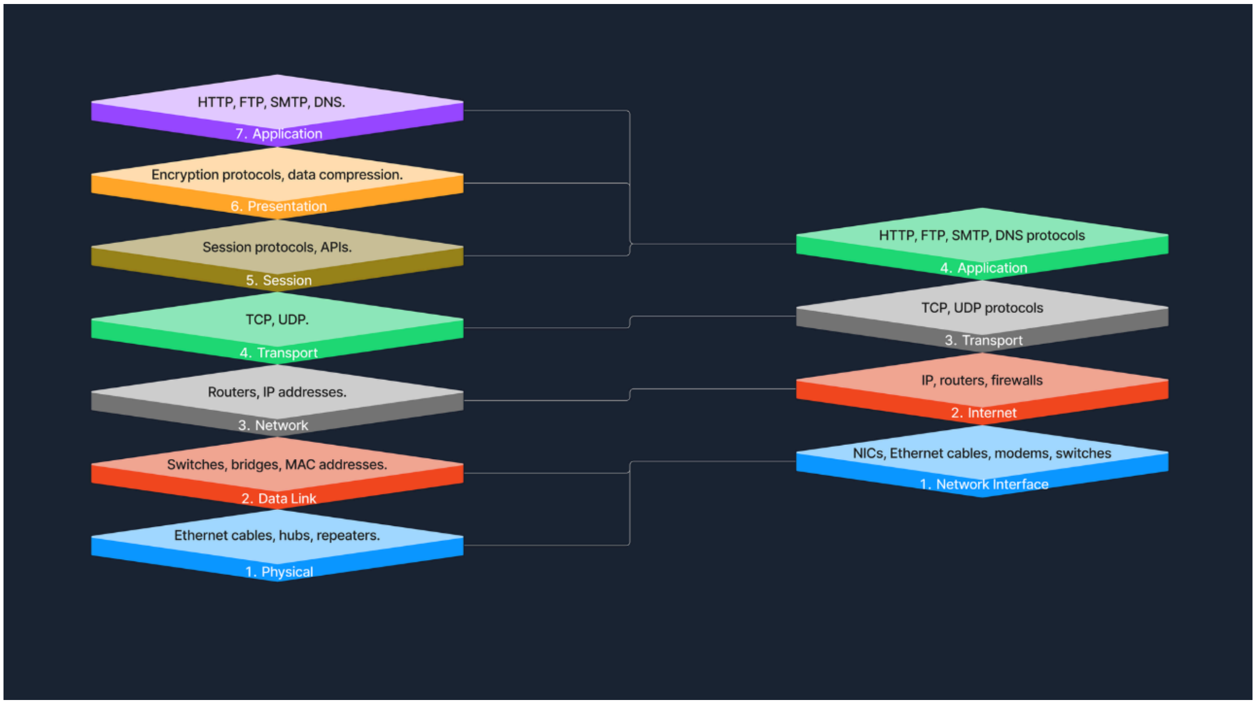
At the `Transport Layer`, the TCP/IP model provides end-to-end communication services that are essential for the functioning of the internet. This includes the use of TCP (Transmission Control Protocol) for reliable communication and UDP (User Datagram Protocol) for faster, connectionless services. This layer ensures that data packets are delivered in a sequential and error-free manner, corresponding to the Transport Layer of the OSI model.

Application Layer

The `Application Layer` of the TCP/IP model contains protocols that offer specific data communication services to applications. Protocols such as HTTP (Hypertext Transfer Protocol), FTP (File Transfer Protocol), and SMTP (Simple Mail Transfer Protocol) enable functionalities like web browsing, file transfers, and email services. This layer corresponds to the top three layers of the OSI model (Session, Presentation, and Application), providing interfaces and protocols necessary for data exchange between systems.



Comparison with OSI Model:



Example of Accessing a Website

When accessing a website, several layers of the TCP/IP model work together to facilitate the process. At the Application Layer, your browser utilizes HTTP to request the webpage. This request then moves to the Transport Layer, where TCP ensures the data is transferred reliably. The Internet Layer comes into play next, with IP taking charge of routing the data packets from our device to the web server. Finally, at the Link Layer, the data is physically transmitted over the network, completing the connection that allows us to view the website.

Model Roles

In practical terms, the TCP/IP model is the backbone of network data transmission, actively employed across various networking environments. On the other hand, the OSI model, while not directly implemented, plays a crucial role as a comprehensive theoretical framework. It helps demystify the complexities of network operations, providing clear insights and a structured approach to understanding how networks function. Together, these models form a complete picture, bridging the gap between theoretical knowledge and practical application in networking.

Protocols

`Protocols` are standardized rules that determine the formatting and processing of data to facilitate communication between devices in a network. These protocols operate at different layers within network models, each tailored to handle specific types of data and communication needs. Here's a look at some common network protocols and their roles in data exchange.

Protocol	Layer	Purpose
HTTP (HyperText Transfer Protocol)	Application (L7)	Transfers web pages between browsers and web servers.
FTP (File Transfer Protocol)	Application (L7)	Transfers files between systems.
SMTP (Simple Mail Transfer Protocol)	Application (L7)	Sends emails between mail servers.
TCP (Transmission Control Protocol)	Transport (L4)	Provides reliable communication with error checking and recovery.
UDP (User Datagram Protocol)	Transport (L4)	Provides fast communication without error recovery.
IP (Internet Protocol)	Network (L3)	Handles addressing and routing of packets between networks.

Transmission

Transmission is the process of sending data from one device to another through a communication medium.

Transmission Types

Analog Transmission

- Uses continuous signals.
- Example: Traditional radio broadcasts.

Digital Transmission

- Uses binary signals (0s and 1s).
- Example: Computer networks, Internet.

Transmission Modes

Simplex

- One-way communication only.
- Example: Keyboard → Computer.

Half-Duplex

- Two-way communication, but one device transmits at a time.
- Example: Walkie-talkies.

Full-Duplex

- Two-way communication simultaneously.
- Example: Phone calls.

Transmission Media

Wired Media

Twisted Pair Cable

- Used in Ethernet and LANs.

Coaxial Cable

- Used in cable TV and older networks.

Fiber Optic Cable

- Uses light signals.
- Very high speed and long distance.

Wireless Media

Radio Waves

- Used in Wi-Fi and cellular networks.

Microwaves

- Used in satellite communication.

Infrared

- Used in remote controls and short-range communication.

Components of a Network

End Devices (Hosts)

Devices that send or receive data.

Examples:

- Computer
- Smartphone
- Tablet
- Smart TV
- IoT Devices

Purpose: User interaction with the network.

Intermediary Devices

Devices that help data travel between networks and devices.

Examples:

- Router
- Switch
- Modem
- Access Point

Purpose: Forward, manage, and secure network traffic.

NIC (Network Interface Card)

Hardware that allows a device to connect to a network.

Types:

- Ethernet NIC (Wired)
- Wi-Fi NIC (Wireless)

Important: Every NIC has a unique **MAC Address**.

Router

- Operates at **Layer 3 (Network Layer)**.
- Uses **IP Addresses**.
- Connects different networks.
- Directs internet traffic.

Example: Home router connecting devices to the Internet.

Switch

- Operates at **Layer 2 (Data Link Layer)**.
- Uses **MAC Addresses**.
- Connects devices within a LAN.
- Sends data only to the intended device.

Hub

- Operates at **Layer 1 (Physical Layer)**.
- Broadcasts data to all connected devices.
- Older and less efficient than switches.

Network Media

Physical or wireless path used for communication.

Wired:

- Ethernet Cable
- Fiber Optic Cable

Wireless:

- Wi-Fi
- Bluetooth

Network Protocols

Rules that allow devices to communicate.

Examples:

- TCP/IP
- HTTP/HTTPS
- FTP
- SMTP

Functions:

- Addressing
- Routing
- Error Checking
- Data Transfer

Network Management Software

Used by administrators to:

- Monitor networks
- Configure devices
- Troubleshoot issues
- Enforce security policies

Software Firewall

Host-based security application.

Functions:

- Monitors network traffic
- Blocks unauthorized access
- Filters malicious packets

Example: Windows Defender Firewall, IPTables

Servers

Computers that provide services to clients.

Types:

- Web Server
- File Server
- Mail Server
- Database Server

Purpose: Store data and provide network services.

> **Which protocol manages data routing and delivery across networks? : TCP/IP**

> **What type of cable is used to connect components within a local area network for high-speed data transfer? : Ethernet**

> **Which device connects multiple networks and manages data traffic to optimize performance? : Router**

> **What type of network cable is used to transmit data over long distances with minimal signal loss? : Fiber-optic cable**

Network Communication

Network communication relies on **3 key components**:

```
MAC Address → Device Identification
IP Address  → Device Location
Port Number → Service/Application Identification
```

MAC Address (Layer 2)

MAC (Media Access Control) Address

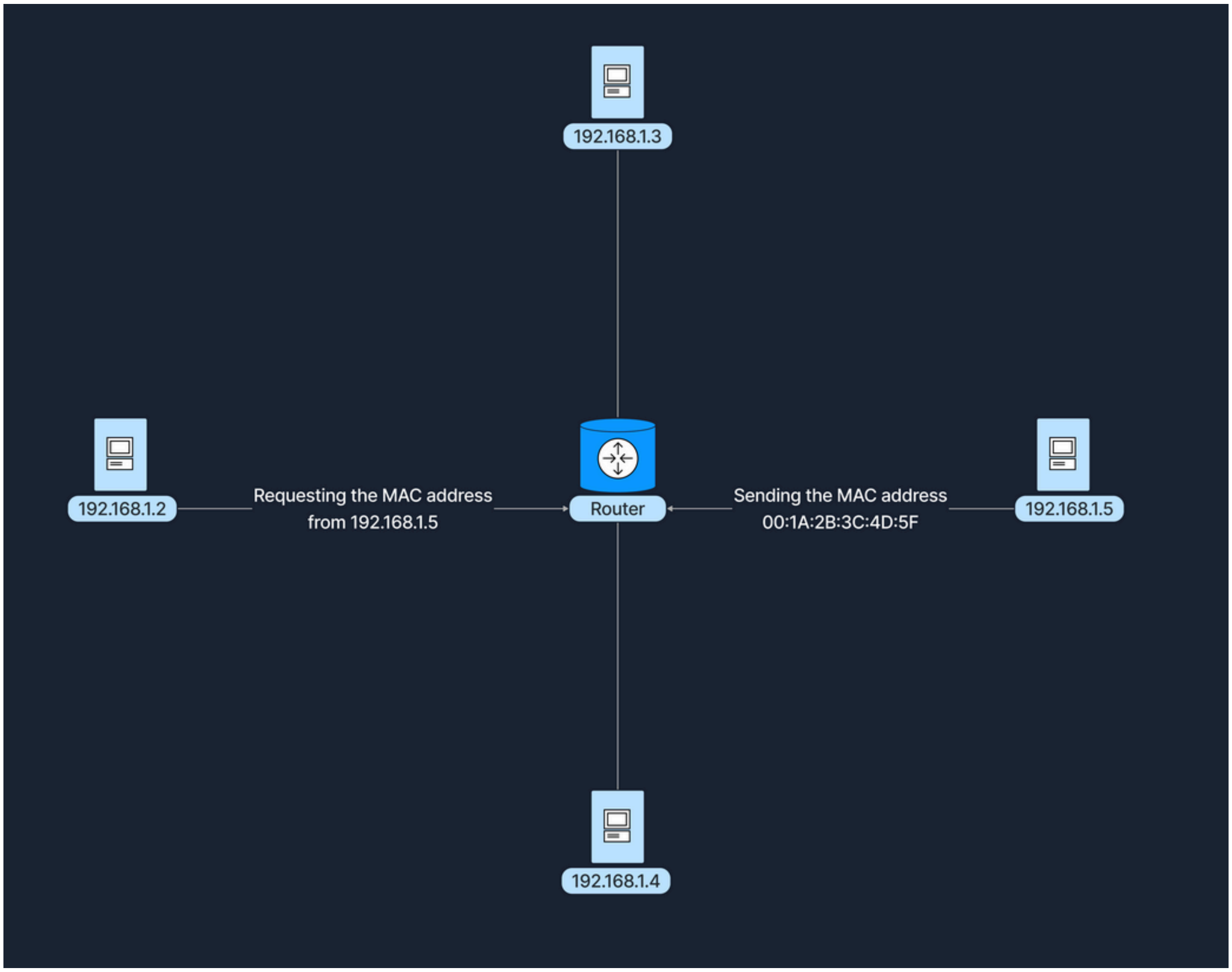
- Unique identifier of a NIC.
- Works in a LAN.
- Used by switches.
- 48-bit hexadecimal address.

Example:

```
00:1A:2B:3C:4D:5E
```

Purpose:

```
Identifies a physical device on a local network.
```



IP Address (Layer 3)

IP (Internet Protocol) Address

- Identifies the location of a device.
- Used by routers.
- Enables communication between networks.

IPv4 Example (32 bits)

```
192.168.1.10
```

IPv6 Example (128 bits)

```
001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Purpose:

```
Identifies where a device is located on a network.
```

Ports (Layer 4)

A port identifies a specific service or application running on a device.

Purpose:

```
IP = House Address
Port = Room Number
```

Example:

```
192.168.1.10:80
```

Device:

```
192.168.1.10
```

Service:

```
HTTP (Port 80)
```

Port Ranges

Range	Type
0-1023	Well-Known Ports

Range	Type
1024-49151	Registered Ports
49152-65535	Dynamic/Private Ports

Common Ports

Port	Service
20/21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
143	IMAP
443	HTTPS
1433	MSSQL

ARP

ARP (Address Resolution Protocol)

Purpose:

```
IP Address → MAC Address
```

Example:

```
192.168.1.5
      ↓
00:1A:2B:3C:4D:5F
```

Web Browsing Process

1. DNS Lookup

Our computer resolves the domain name to an IP address (e.g., `93.184.216.34` for `example.com`).

2. Data Encapsulation

Step
Your browser generates an HTTP request.

The request is encapsulated with TCP, specifying the destination port
`80`
or
`443`

The packet includes the destination IP address `93.184.216.34`
On the local network, our computer uses ARP to find the MAC address of the default gateway (router).

3. Data Transmission

Step
The data frame is sent to the router's MAC address.

The router forwards the packet toward the destination IP address.
Intermediate routers continue forwarding the packet based on the IP address.

4. Server Processing

Step
The server receives the packet and directs it to the application listening on port

```
80
```

or

```
443
```

The server processes the HTTP request and sends back a response following the same path in reverse.

5. Response Transmission

Step
The server sends the response back to the client’s temporary port, which was randomly selected by the client’s operating system at the start of the session.
The response follows the reverse path back through the network, being directed from router to router based on the source IP address and port information until it reaches the client.

> **Domain**
> ↓
> **DNS**
> ↓

> **IP Address**
> **↓**
> **TCP Connection**
> **↓**
> **HTTPS Request**
> **↓**
> **Server Response**
> **↓**
> **Web Page Appears**

Tools: getmac (for mac address)
: netstat (to view active connection and listening) (eg. netstat -ano -p tcp)

DHCP (Dynamic Host Configuration Protocol)

What is DHCP?

DHCP automatically assigns network settings to devices when they join a network.

Instead of manually setting:

```
IP Address
Subnet Mask
Default Gateway
DNS Server
```

DHCP does it automatically.

Why DHCP?

Without DHCP

```
Admin manually assigns IPs.
Risk of duplicate IPs.
Time consuming.
```

With DHCP

```
Automatic IP assignment.
No IP conflicts.
Easy network management.
```

DHCP Components

Role	Description
DHCP Server	Assigns IP addresses and network settings.
DHCP Client	Device requesting an IP address.

Example

```
Router = DHCP Server

Laptop
Phone
Tablet
= DHCP Clients
```

DORA Process

DHCP works using **DORA**:

Step	Meaning
D	Discover
O	Offer
R	Request
A	Acknowledge

---

1. Discover

Client broadcasts:

```
"Any DHCP server available?"
```

---

2. Offer

DHCP Server replies:

```
"I can give you IP 192.168.1.10"
```

---

3. Request

Client says:

```
"I want 192.168.1.10"
```

---

4. Acknowledge

Server confirms:

```
"IP assigned successfully"
```

Lease Time

DHCP does not permanently assign IPs.

Example:

```
IP: 192.168.1.10
Lease: 24 Hours
```

Before expiration:

```
Client requests renewal.
Server extends lease.
```

Real Example

Alice connects her laptop:

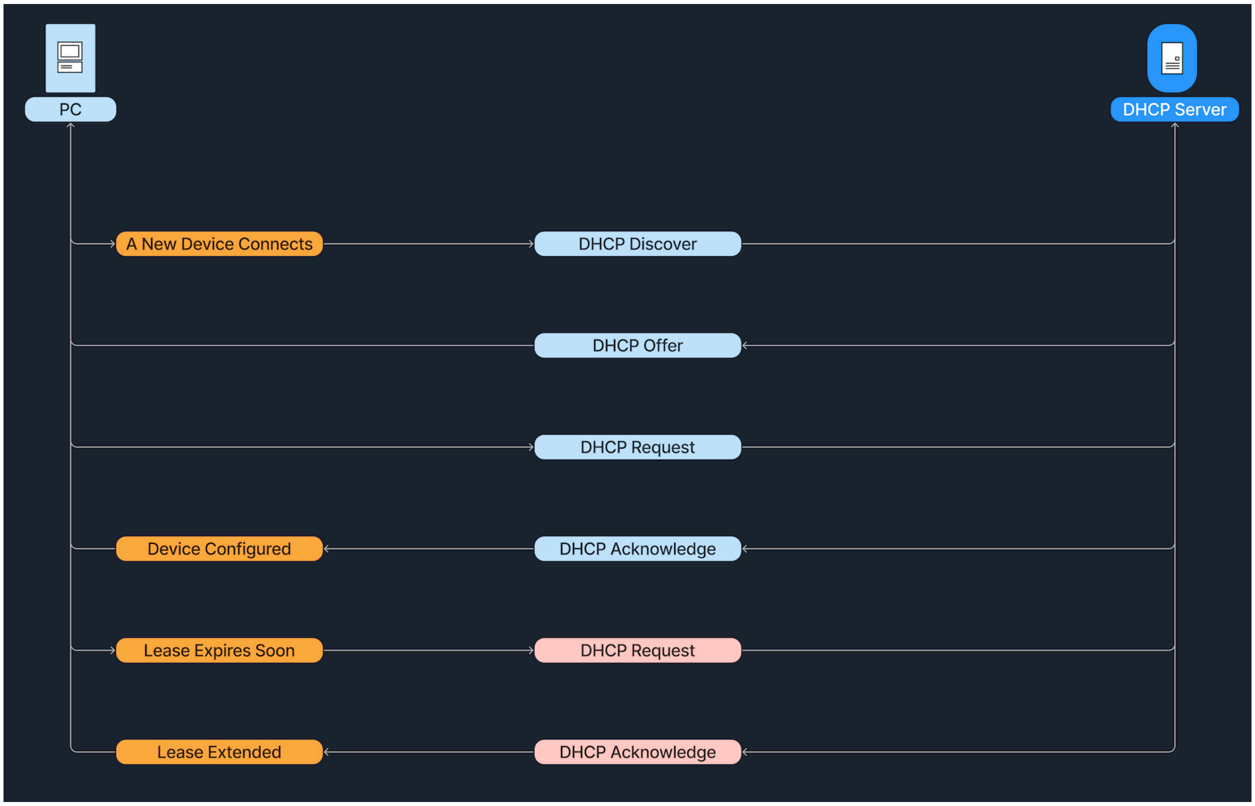
```
Laptop
 ↓
Discover

DHCP Server
 ↓
Offer 192.168.1.10

Laptop
 ↓
Request

DHCP Server
 ↓
Acknowledge

Laptop receives IP
```



eg. sudo dhclient wlan0 (check packet on wireshark)

NAT (Network Address Translation)

Why NAT Exists

IPv4 has limited addresses:

```
~4.3 Billion IPs
```

NAT allows:

```
Many Devices
      ↓
One Public IP
```

Public vs Private IP

Public IP

- Unique on the Internet.
- Assigned by ISP.
- Reachable from anywhere.

Example:

```
8.8.8.8
```

Private IP

- Used inside local networks.
- Not reachable directly from the Internet.

Ranges:

```
10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

192.168.0.0 - 192.168.255.255
```

Example:

```
192.168.1.10
```

What NAT Does

Example:

```
Laptop  = 192.168.1.10
Phone   = 192.168.1.11
```

```
PC      = 192.168.1.12

Router Public IP
= 203.0.113.50
```

When laptop visits Google:

```
Before NAT

192.168.1.10
     ↓
Google
```

Router changes it to:

```
After NAT

203.0.113.50
     ↓
Google
```

Google thinks:

```
Request came from 203.0.113.50
```

How Reply Comes Back

Router keeps a NAT table:

```
203.0.113.50:4444
        ↓
192.168.1.10:5555
```

When Google replies:

```
203.0.113.50
```

Router checks table and sends it to:

```
192.168.1.10
```

Types of NAT

Static NAT

Involves a one-to-one mapping, where each private IP address corresponds directly to a public IP address.

```
1 Private IP
      ↔
1 Public IP
```

Example:

- Web Servers
- Mail Servers
- VPN Servers
- DNS Servers

```
192.168.1.10
      ↔
203.0.113.10
```

Dynamic NAT

Assigns a public IP from a pool of available addresses to a private IP as needed, based on network demand.

```
Public IP assigned from a pool.
```

Example:

```
Pool:
203.0.113.10
203.0.113.11
203.0.113.12
```

PAT (Port Address Translation)

Also called:

```
NAT Overload
```

Multiple devices share one public IP using different ports.

Example:

192.168.1.10 → 203.0.113.50:4001
192.168.1.11 → 203.0.113.50:4002
192.168.1.12 → 203.0.113.50:4003

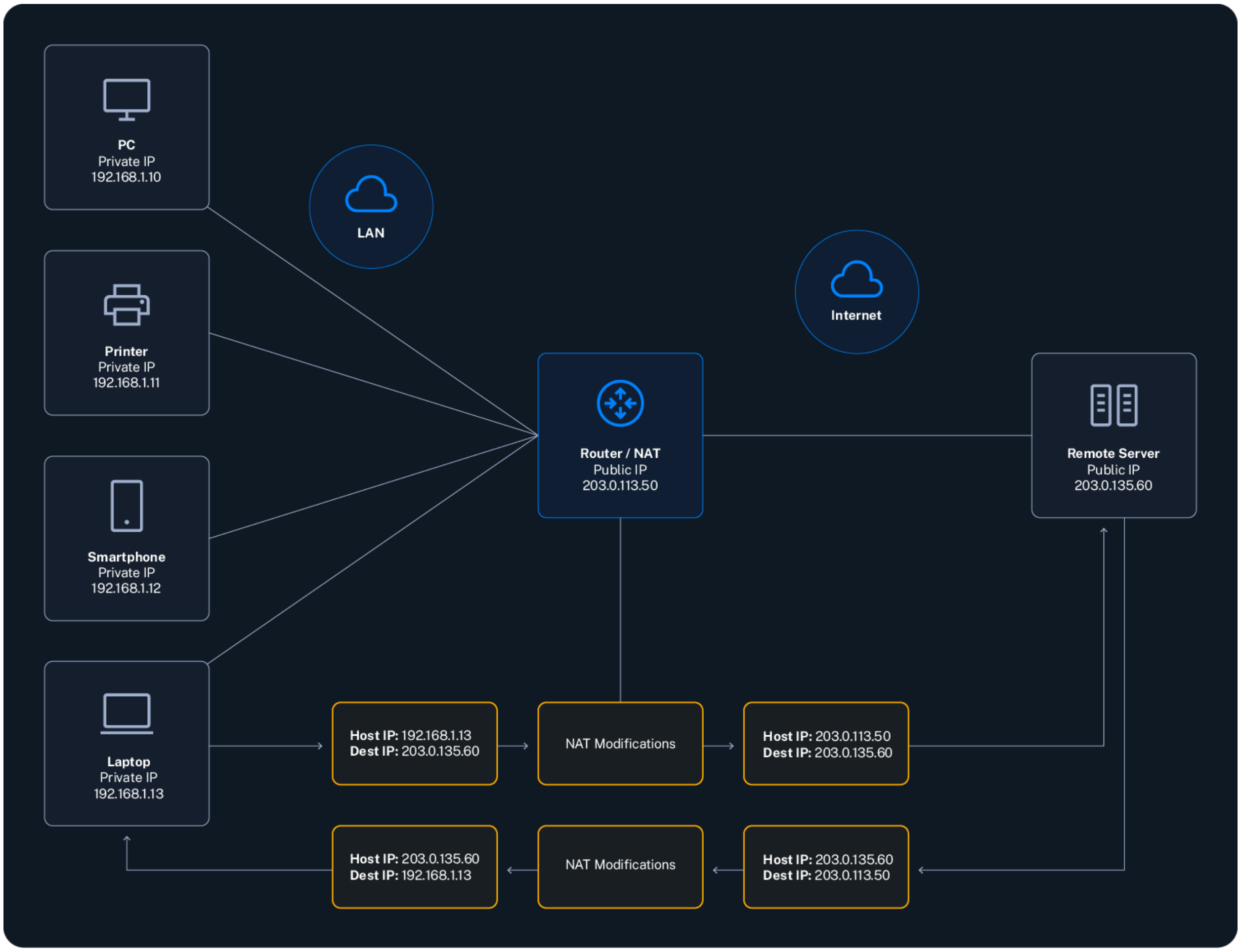
This is what almost every home router does.

Benefits

- Saves public IP addresses
- Hides internal devices
- Easy home networking

Drawbacks

- Hosting servers is harder
- May require Port Forwarding
- Makes troubleshooting harder



In big company:

- **Static NAT** is for important servers that must always be reachable.
- **Dynamic NAT** is for sharing a pool of public IPs.
- **PAT** is still doing most of the heavy lifting for employee Internet access.

DNS (Domain Name System)

What is DNS?

DNS is the **phonebook of the Internet**.

It converts:

```
Domain Name → IP Address
```

Example:

```
google.com
     ↓
142.250.x.x
```

Without DNS, we would have to remember IP addresses instead of website names.

Domain Name vs IP Address

Type	Example
Domain Name	google.com
IP Address	142.250.190.78

DNS Hierarchy

```
Root (.)
   ↓
TLD (.com, .org, .net)
   ↓
Second Level Domain (google)
   ↓
Subdomain (www)
```

Example:

```
www.google.com

www      = Subdomain
google   = Second-Level Domain
.com     = Top-Level Domain (TLD)
```

DNS Resolution Process

Step	What Happens
1	User enters **www.example.com** in the browser.
2	Computer checks the **local DNS cache** for the IP address.
3	If not found, it queries a **Recursive DNS Server** (ISP DNS, Google DNS, etc.).
4	The Recursive DNS Server contacts a **Root DNS Server**.
5	The Root Server points to the correct **TLD Server** (.com, .org, .net).
6	The TLD Server points to the **Authoritative DNS Server** for the domain.
7	The Authoritative DNS Server returns the domain's **IP Address**.
8	The Recursive DNS Server sends the IP Address back to the computer.
9	The browser connects to the website using that IP Address.

DNS Flow

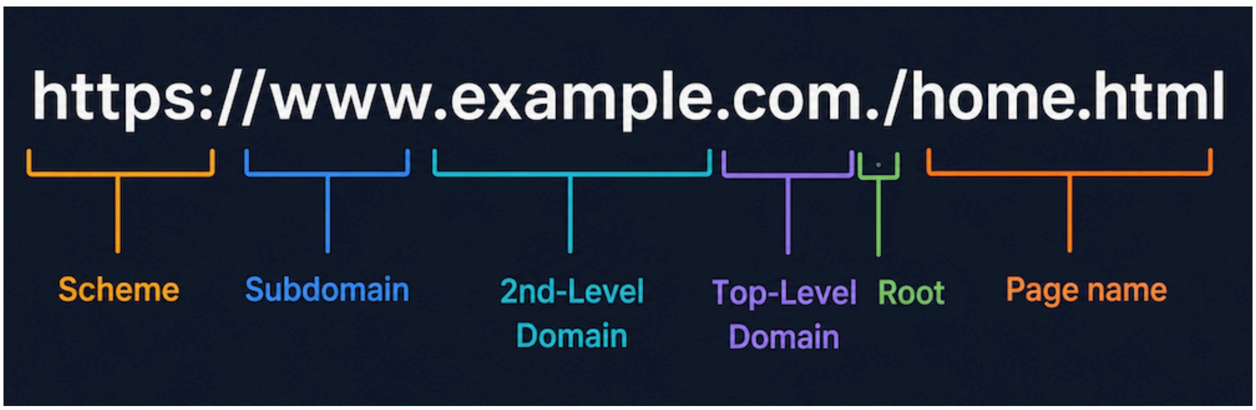
```
google.com
     ↓
DNS Cache
     ↓
Recursive DNS
     ↓
Root Server
     ↓
TLD Server
     ↓
Authoritative Server
     ↓
IP Address
```

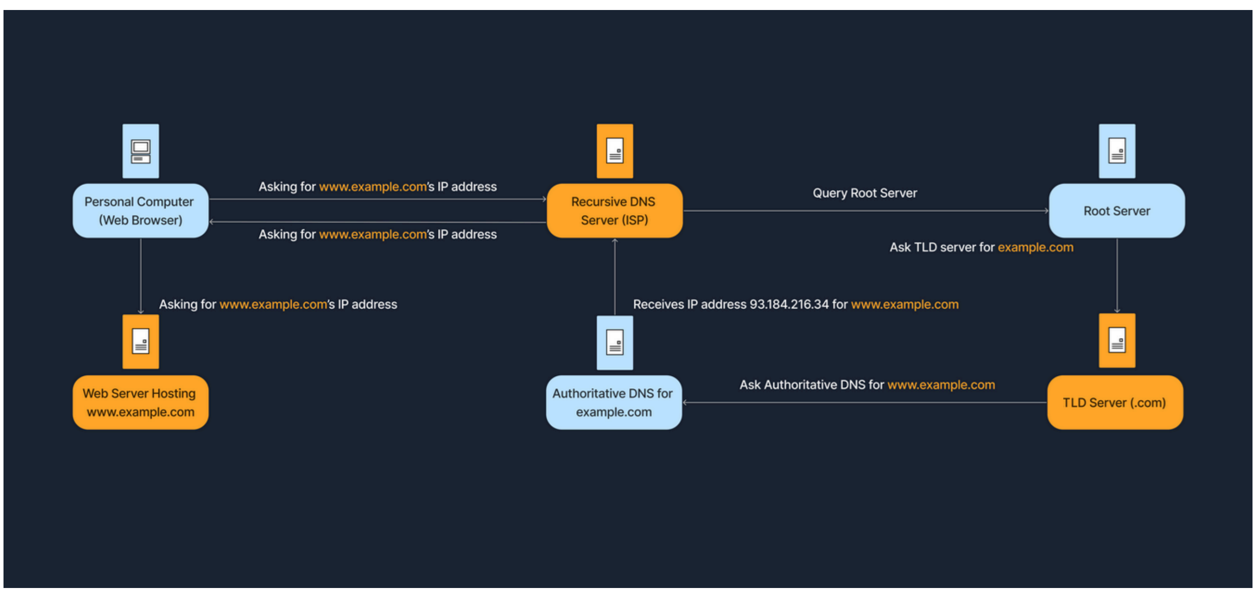
```
     ↓
Website Opens
```

Common DNS Servers

```
Google DNS
8.8.8.8
8.8.4.4

Cloudflare DNS
1.1.1.1
1.0.0.1
```





Internet Architectures

Peer-to-Peer (P2P)

Every device acts as both a client and a server. Devices communicate directly without a central server.

Example

- BitTorrent
- File Sharing

Advantages

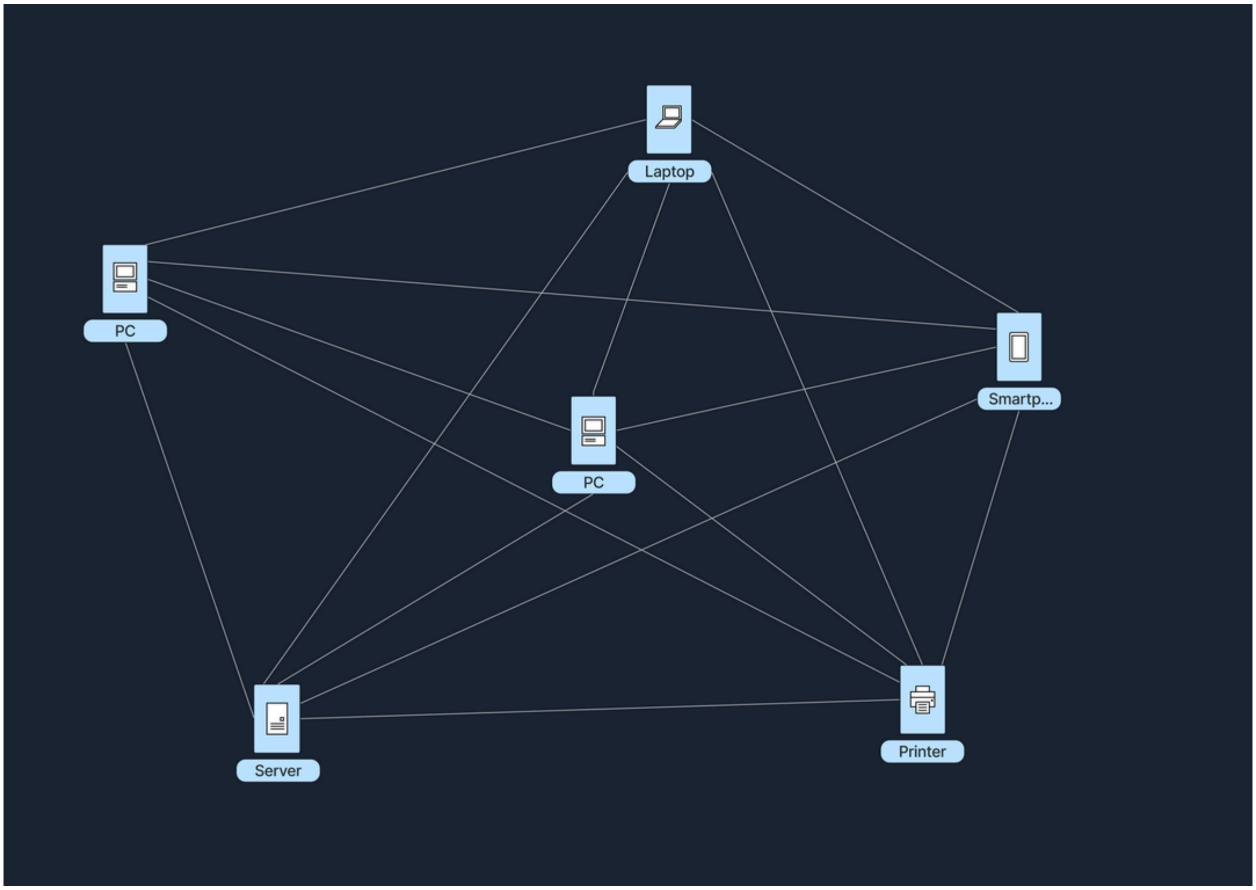
- Highly scalable
- No single point of failure
- Resource sharing among peers

Disadvantages

- Hard to manage
- Security risks
- Resources unavailable if peers disconnect

Memory

```
P2P = Everyone Shares
```



Client-Server Architecture

Clients send requests, and a central server responds.

Example

- Websites
- Email Services
- Online Banking

Advantages

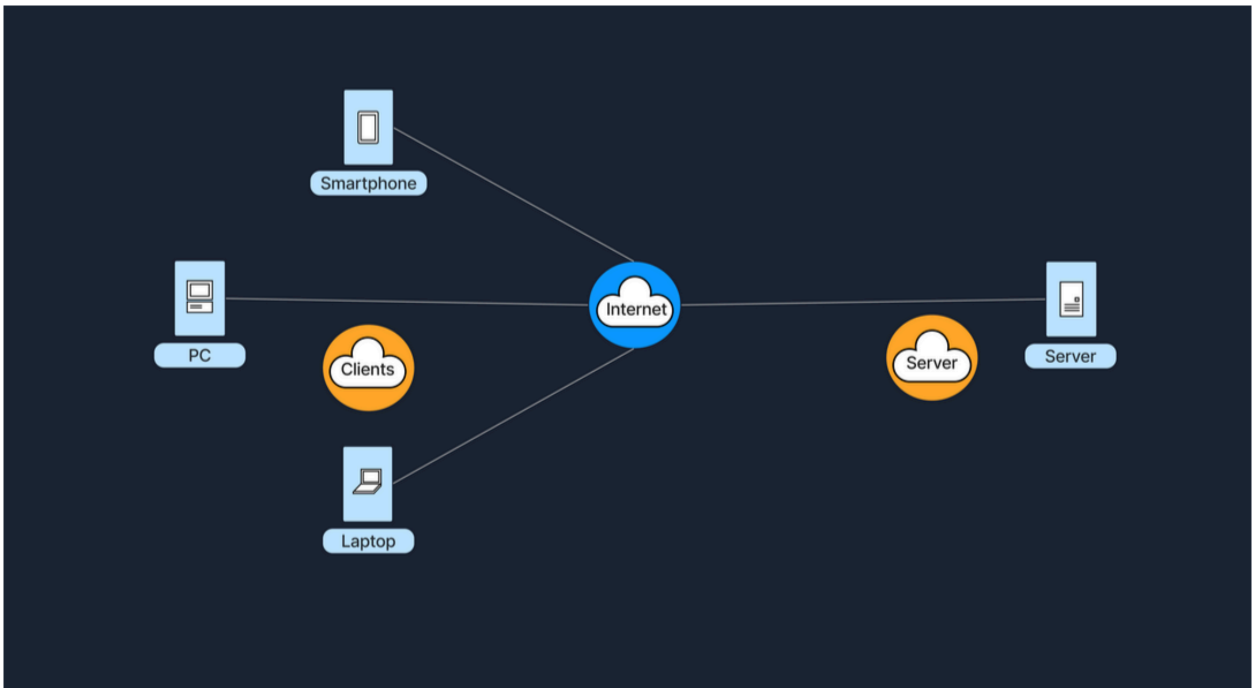
- Centralized control
- Easier management
- Better security

Disadvantages

- Single point of failure
- Expensive infrastructure
- Network congestion

Memory

```
Client Requests → Server Responds
```



Single-Tier Architecture

Client, application, and database are on the same machine.

Example

- Simple desktop application

Memory

```
Everything on One System
```

Two-Tier Architecture

Client communicates directly with a database server.

Example

- Desktop application connected to SQL database

Memory

```
Client ↔ Database
```

Three-Tier Architecture

Adds an application server between client and database.

Example

```
Browser
   ↓
Web Server
   ↓
Database
```

Memory

```
Client ↔ Application ↔ Database
```

N-Tier Architecture

Uses multiple application layers for large-scale applications.

Example

- Amazon
- Netflix
- Enterprise Applications

Memory

```
Many Layers = Better Scalability
```

Hybrid Architecture

Combination of Client-Server and P2P.

Example

- Video Conferencing
- Messaging Apps

Advantages

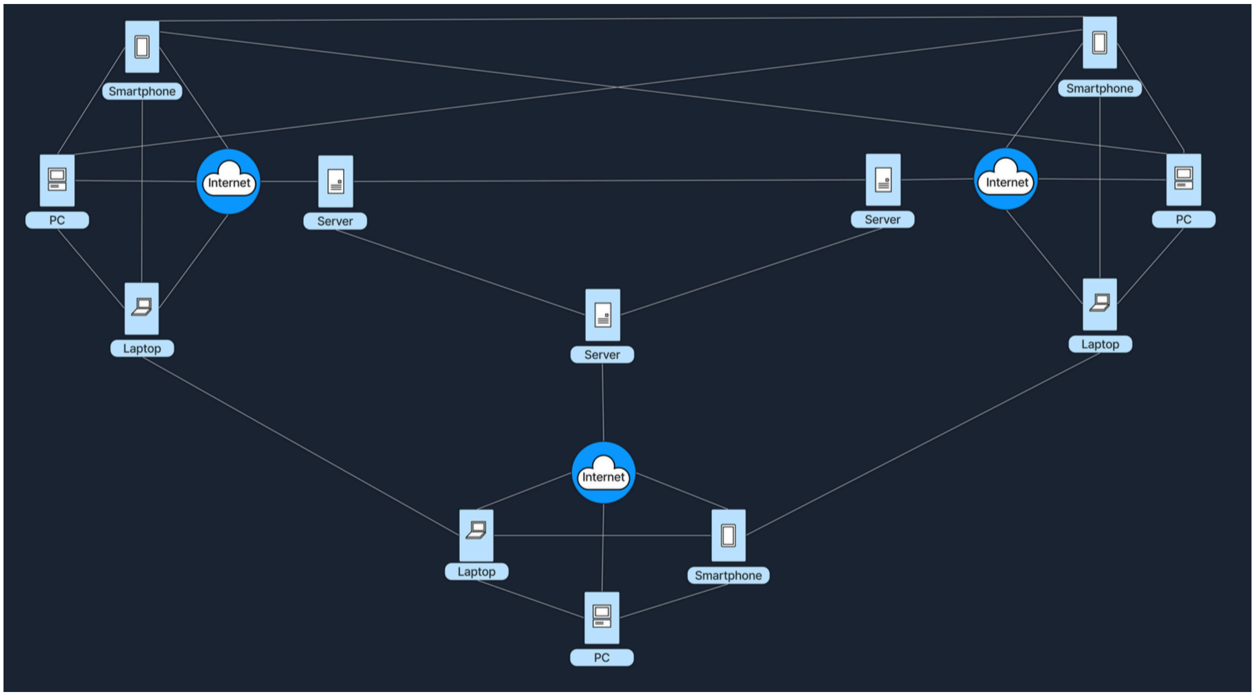
- Efficient
- Less server load

Disadvantages

- More complex
- Central server may still fail

Memory

```
Hybrid = Best of Both Worlds
```



Cloud Architecture

Resources are hosted by cloud providers and accessed over the Internet.

Example

- AWS
- Azure
- Google Cloud
- Google Drive
- Dropbox

Advantages

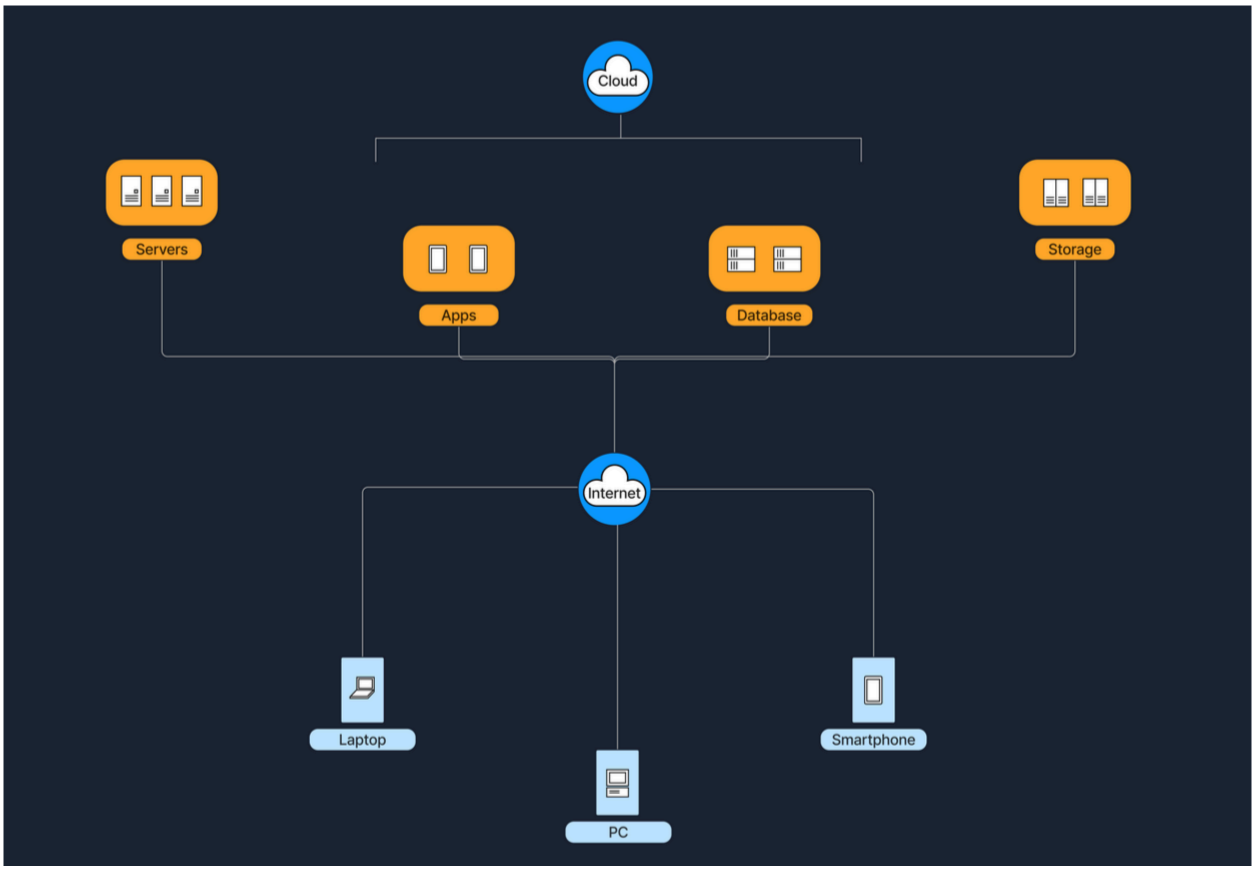
- Scalable
- Reduced maintenance
- Access from anywhere

Disadvantages

- Internet required
- Vendor lock-in
- Privacy concerns

Memory

```
Cloud = Someone Else's Computer
```



Software-Defined Architecture (SDN)

Software-Defined Networking (SDN)

Separates network control from forwarding devices.

Example

- Data Centers

- Cloud Providers

Advantages

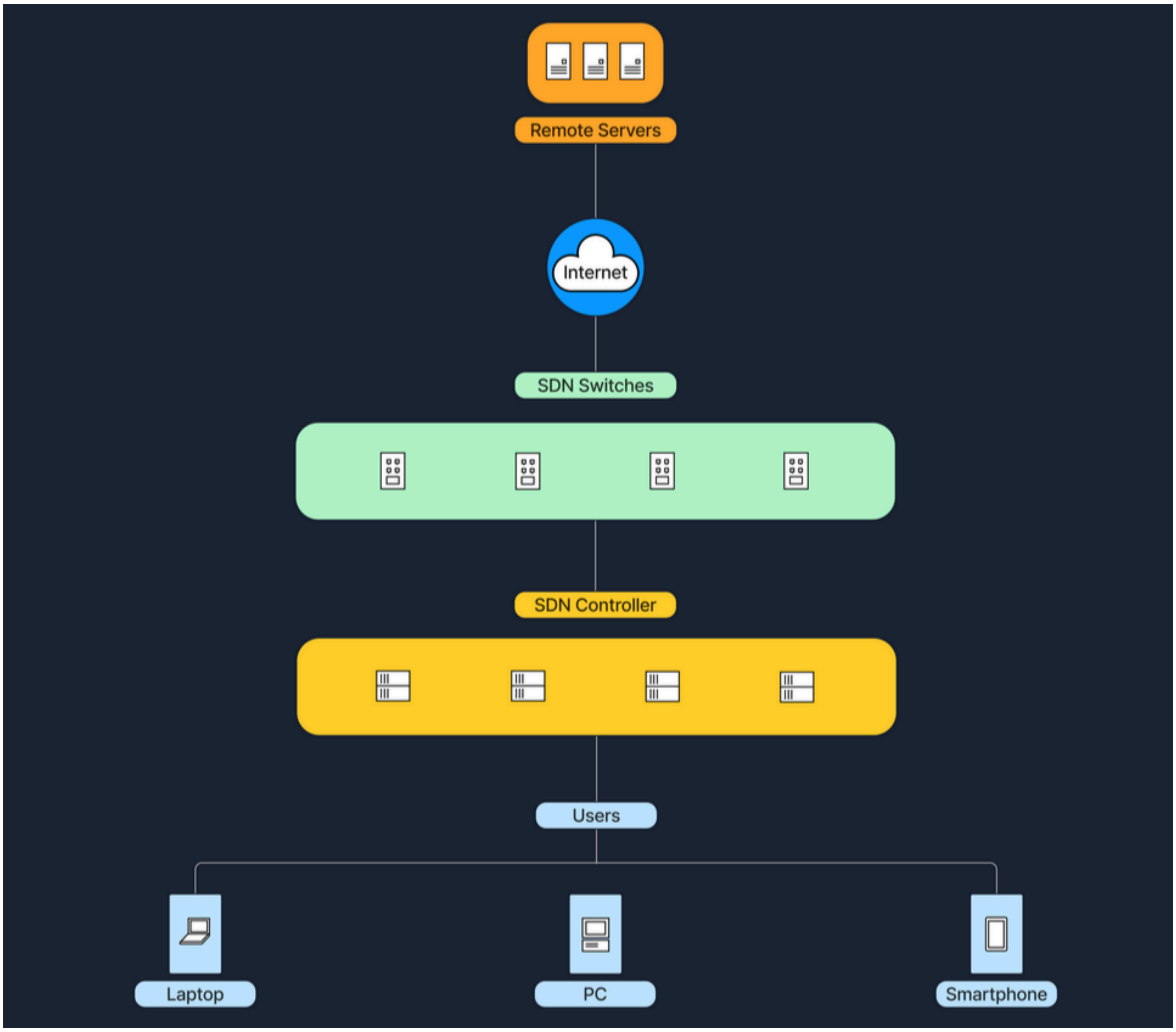
- Centralized management
- Automation
- Flexible networking

Disadvantages

- Controller failure risk
- Complex implementation

Memory

```
SDN = Network Controlled by Software
```



Architecture	Centralized	Scalability	Management	Typical Use Cases
P2P	Decentralized	High	Complex	Torrenting, Blockchain
Client-Server	Centralized	Moderate	Easy	Websites, Email Services
Hybrid	Partially Centralized	High	Moderate	Zoom, Teams, Messaging Apps

Architecture	Centralized	Scalability	Management	Typical Use Cases
Cloud	Centralized (Provider Managed)	Very High	Easy	AWS, Google Drive, SaaS
SDN	Centralized Control Plane	High	Moderate	Datacenters, Large Enterprises